Quiz 2

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Contents

Please note the quiz is based on the following topics:

- Subgroups.
- Cyclic groups.

Problem 1:
Which of the following elements generate the group $\mathbb{Z}_3 \times \mathbb{Z}_4$?

(a) $(1, 2)$.

Solution:

$(1, 2)$ doesn't generate $\mathbb{Z}_3 \times \mathbb{Z}_4$ since:

$$\begin{aligned}\langle(1,2)\rangle &= \{\, 0(1,2), 1(1,2), 2(1,2), 3(1,2), 4(1,2), 5(1,2) \,\} \\ &= \{\, (0,0), (1,2), (2,0), (0,2), (1,0), (2,2) \,\}\end{aligned}$$

and because an element like $(2, 1)$ belongs to $\mathbb{Z}_3 \times \mathbb{Z}_4$ but doesn't belong to $\langle(1, 2)\rangle$, $(1, 2)$ can't be a generator for $\mathbb{Z}_3 \times \mathbb{Z}_4$.

(b) $(1, 1)$.

Solution:

$(1, 1)$ does indeed generate $\mathbb{Z}_3 \times \mathbb{Z}_4$. To see why, we can try a different approach than the one used for **(a)**.

It is clearly the case that the element 1 in $\mathbb{Z}_3$ has order 3 and $1 \in \mathbb{Z}_4$ has order 4 (since $(3)(1) \pmod 3 \equiv 0$ and $(4)(1) \pmod 4 \equiv 0$ respectively). By an earlier theorem, we know that the order of $(1, 1)$ is therefore given by:

$$\begin{aligned}\operatorname{ord}((1,1)) &= \operatorname{lcm}(3,4) \\ &= 12\end{aligned}$$

And since $|\mathbb{Z}_3 \times \mathbb{Z}_4| = |\mathbb{Z}_3||\mathbb{Z}_4| = (3)(4) = 12$ as well, $(1, 1)$ has the same order as $\mathbb{Z}_3 \times \mathbb{Z}_4$ which means that $(1, 1)$ generates $\mathbb{Z}_3 \times \mathbb{Z}_4$!

(c) $(2, 3)$.

Solution:

$(2, 3)$ does indeed generate $\mathbb{Z}_3 \times \mathbb{Z}_4$. Using a brute-force approach, as we did in **(a)**, we have that:

$$\begin{array}{lll} 0(2,3) = (0,0) & 4(2,3) = (2,0) & 8(2,3) = (1,0) \\ 1(2,3) = (2,3) & 5(2,3) = (1,3) & 9(2,3) = (0,3) \\ 2(2,3) = (1,2) & 6(2,3) = (0,2) & 10(2,3) = (2,2) \\ 3(2,3) = (0,1) & 7(2,3) = (2,1) & 11(2,3) = (1,1) \end{array}$$

And so, since $\langle(2, 3)\rangle = \{\, (0, 0), (0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 1), (2, 2), (2, 3) \,\} = \mathbb{Z}_3 \times \mathbb{Z}_4$, $(2, 3)$ is a generator for $\mathbb{Z}_3 \times \mathbb{Z}_4$.

(d) $(0, 1)$.

Solution:

$(0, 1)$ doesn't generate $\mathbb{Z}_3 \times \mathbb{Z}_4$. To see why, all we need is recognize that the element $(1, 1)$ belongs to $\mathbb{Z}_3 \times \mathbb{Z}_4$.

For $(0, 1)$ to actually generate $\mathbb{Z}_3 \times \mathbb{Z}_4$, there must exist some integer $k \in \mathbb{Z}$ such that $k(0, 1) = (1, 1)$. However, this cannot be the case since such a k would necessarily have to satisfy $0k = 1$, which is clearly impossible!

Thus, $\langle (0, 1) \rangle$ cannot possibly generate $\mathbb{Z}_3 \times \mathbb{Z}_4$.

Problem 2:
In the group $\mathbb{Z}_5 \times \mathbb{Z}_6$, what is the order of the element $(2,3)$?

Solution:

The order of $(2,3)$ in $\mathbb{Z}_5 \times \mathbb{Z}_6$ is 10.

To find the order of $(2,3)$ in $\mathbb{Z}_5 \times \mathbb{Z}_6$, we can leverage the fact that, since $\mathbb{Z}_5 \times \mathbb{Z}_6$ is the group formed by the direct product of the groups $\mathbb{Z}_5$ and $\mathbb{Z}_6$, the order of any element $(a,b) \in \mathbb{Z}_5 \times \mathbb{Z}_6$ is the lowest common multiple of $\operatorname{ord}(a)$ and $\operatorname{ord}(b)$.

Thus, $\operatorname{ord}((2,3)) = \operatorname{lcm}(2,3)$, where $2 \in \mathbb{Z}_5$ and $3 \in \mathbb{Z}_6$ respectively. Now, because $\operatorname{ord}(2) = 5$ and $\operatorname{ord}(3) = 2$ in their respective groups (which can be easily verified in the computations below):

$$\begin{aligned} (1)(2) &\equiv 2 \\ (2)(2) &\equiv 4 \\ (3)(2) &\equiv 1 \\ (4)(2) &\equiv 3 \\ (5)(2) &\equiv 0 \end{aligned} \qquad\qquad \begin{aligned} (1)(3) &\equiv 3 \\ (2)(3) &\equiv 0 \end{aligned}$$

we have that $\operatorname{ord}((2,3)) = \operatorname{lcm}(5,2) = 10$.

Problem 3:
What are the orders of the following permutations?

(a) (1234).

Solution:

Brute forcing, we have that $\operatorname{ord}((1234)) = 4$ since:

n	$(1234)^{n-1}$	$\circ$	(1234)	$=$	$(1234)^n$
1	$()$	$\circ$	(1234)	$=$	(1234)
2	(1234)	$\circ$	(1234)	$=$	$(13)(24)$
3	$(13)(24)$	$\circ$	(1234)	$=$	(1432)
4	(1432)	$\circ$	(1234)	$=$	$()$

(b) (123)(456).

Solution:

Brute forcing once again, we have that $\operatorname{ord}((123)(456)) = 3$ since:

n	$((123)(456))^{n-1}$	$\circ$	$(123)(456)$	$=$	$((123)(456))^n$
1	$()$	$\circ$	$(123)(456)$	$=$	$(123)(456)$
2	$(123)(456)$	$\circ$	$(123)(456)$	$=$	$(132)(465)$
3	$(132)(456)$	$\circ$	$(123)(456)$	$=$	$()$

Notice as well that we could have also easily deduced that $\operatorname{ord}((123)(456)) = 3$ by the fact that $\operatorname{ord}((123)) = \operatorname{ord}((456)) = 3$, and so since $\operatorname{ord}((123)(456)) = \operatorname{lcm}(\operatorname{ord}((123)), \operatorname{ord}((456))) = \operatorname{lcm}(3,3)$, we'd automatically have that $\operatorname{ord}((123)(456)) = 3$.

(c) (12)(345).

Solution:

$\operatorname{ord}((12)(345)) = 6$ because, by the following tables:

n	$(12)^{n-1}$	$\circ$	(12)	$=$	$(12)^n$
1	$()$	$\circ$	(12)	$=$	(12)
2	(12)	$\circ$	(12)	$=$	$()$

n	$(345)^{n-1}$	$\circ$	(345)	$=$	$(345)^n$
1	$()$	$\circ$	(345)	$=$	(345)
2	(345)	$\circ$	(345)	$=$	(354)
3	(354)	$\circ$	(345)	$=$	$()$

$\operatorname{ord}((12)) = 2$ and $\operatorname{ord}((345)) = 3$, and since $\operatorname{ord}((12)(345)) = \operatorname{lcm}(\operatorname{ord}((12)), \operatorname{ord}((345)))$, we have that:

$$\begin{aligned}\operatorname{ord}((12)(345)) &= \operatorname{lcm}(\operatorname{ord}((12)), \operatorname{ord}((345))) \\ &= \operatorname{lcm}(2,3) \\ &= 6\end{aligned}$$

(d) (12)(3456).

Solution:

Generalizing our earlier findings, we can conclude that $\operatorname{ord}((12)(3456)) = 4$, since $\operatorname{ord}((12)) = 2$ and $\operatorname{ord}((3456)) = 4$, and since $\operatorname{ord}((12)(3456)) = \operatorname{lcm}(\operatorname{ord}((12)), \operatorname{ord}((3456)))$:

$$\begin{aligned}\operatorname{ord}((12)(3456)) &= \operatorname{lcm}(\operatorname{ord}((12)), \operatorname{ord}((3456))) \\ &= \operatorname{lcm}(2, 4) \\ &= 4\end{aligned}$$

Problem 4:
How many element A_5 have order 5?

Solution:

To find the number of elements in A_5 that have order 5, we will (admittedly somewhat unnecessarily) proceed by first proving that a claim that permutation in A_5 has order 5 if and only if its cycle type is (5).

(I) Prove the claim's forward direction.

The forward direction of the claim states that if a permutation $\sigma \in A_5$ has order 5, then it has a cycle type of (5). To prove the claim in this direction, we need to show that any element in the set $\{1, 2, 3, 4, 5\}$ must be permuted by σ exactly a multiple of 5 times before equalling itself. Or, in other words, we need to show that if $a \in \{1, 2, 3, 4, 5\}$, then $\sigma^n(a) = a$ if and only if $n = 5k$ for some integer $k \in \mathbb{Z}$.

We will therefore proceed by proving this subclaim.

(a) Prove the forward direction of the subclaim.

The subclaim's forward direction states that if $\sigma^n(a) = a$ for all $a \in \{1, 2, 3, 4, 5\}$, then there exists some integer $k \in \mathbb{Z}$ such that $n = 5k$.

To see why this is the case, we need only recognize that if $\sigma^n(a) = a$ for all such a, then $\sigma^n(a) = ()$, by definition of the identity permutation. Recalling as well that $\operatorname{ord}(\sigma)$ was assumed to be equal to 5 earlier, we know that $\sigma^5 = ()$ by definition of order. Thus, we know by Theorem 4.3 that 5 must divide n, and hence there must exist some integer $k \in \mathbb{Z}$ such that $n = 5k$, by definition of divisibility by 5.

Thus, since $\sigma^n(a) = a$ for all $a \in \{1, 2, 3, 4, 5\}$ implies $n = 5k$ for all $k \in \mathbb{Z}$, the forward direction of our claim has been proven.

(b) Prove the reverse direction of the subclaim.

The reverse direction of the subclaim states that if $n = 5k$ for some integer $k \in \mathbb{Z}$, then $\sigma^n(a) = a$ for all $a \in \{1, 2, 3, 4, 5\}$.

This is actually, and thankfully, quite easy to show. To see why this direction is true, recall that $\operatorname{ord}(\sigma) = 5$ by assumption and so by definition of order, $\sigma^5 = ()$. Thus, letting $a \in \{1, 2, 3, 4, 5\}$ be arbitrary, we have that:

$$
\begin{aligned}
\sigma^n(a) &= \sigma^{5k}(a) && \text{(By our earlier characterization of } n\text{)} \\
&= (\sigma^5)^k(a) && \text{(By exponent laws)} \\
&= (()^k)(a) && \text{(Since } \operatorname{ord}(\sigma) = 5\text{)} \\
&= ()(a) && \text{(Inductively by definition of the identity)} \\
&= a && \text{(By definition of the identity)}
\end{aligned}
$$

And thus, since $\sigma^n(a) = a$ for all $a \in \{1, 2, 3, 4, 5\}$ whenever $n = 5k$ for some integer $k \in \mathbb{Z}$, the subclaim's reverse direction is proven.

And so, since both **(a)** and **(b)** both hold, the subclaim's if-and-only-if premise logically follows. Hence, because the subclaim implies that $\sigma^n(a) = a$ for all $a \{1, 2, 3, 4, 5\}$ if and only if n is a multiple of 5, it clearly follows that σ must have a cycle type of (5).

Therefore, since σ having an order of 5 implies it has a cycle type of (5), the original claim's forward direction has been successfully proven.

(II) Prove the reverse direction of the claim.

The reverse direction states that if a permutation $\sigma \in A_5$ has a cycle type of (5), then it has an order of 5. To prove the claim in this direction, we need to show that $n = 5$ is the smallest positive integer satisfying $\sigma^n = ()$.

Thankfully, this too is quite each to prove. Because σ has a cycle type of (5) by assumption, we can express that σ is of the form $(abcde)$, where $a, b, c, d, e \in \{1, 2, 3, 4, 5\}$ and each of a, b, c, d, and e are pairwise distinct. So, by brute forcing, we find that:

n	σ^{n-1}	$\circ$	σ	$=$	σ^n
1	$()$	$\circ$	$(abcde)$	$=$	$(abcde)$
2	$(abcde)$	$\circ$	$(abcde)$	$=$	$(acebd)$
3	$(acebd)$	$\circ$	$(abcde)$	$=$	$(adbec)$
4	$(adbec)$	$\circ$	$(abcde)$	$=$	$(aedcb)$
5	$(aedcb)$	$\circ$	$(abcde)$	$=$	$()$

And thus, since σ^5 both equals the identity permutation $()$, and is the smallest positive power of σ to do so, $\operatorname{ord}(\sigma)$ is necessarily 5 by definition.

Thus, since **(I)** and **(II)** have been shown to hold, our claim that $\sigma \in A_5$ has an order of 5 if and only if it has a cycle type of (5) is proven. Phew!

And so, since we know that A_5 has $\frac{5!}{5} = \frac{120}{5} = 24$ permutation of cycle type (5) we can know conclusively say that A_5 has exactly 24 elements with an order of 5 by the claim we just proved.

Problem 5:
In the group $GL_2(\mathbb{Z}_3)$, how many elements are in the subgroup generated by $\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$?

Solution:

Considering how involved our more rigorous earlier proof got, we're just going to solve this one by brute force lol.

Intuitively, the number of elements in the subgroup generated by $\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$ is just the number of distinct powers of $\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$, which is just its order. And since:

n	$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}^{n-1}$	$\cdot$	$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$	$=$	$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}^{n}$
1	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\cdot$	$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$	$=$	$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$
2	$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$	$\cdot$	$\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$	$=$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

we have that $\text{ord}\left(\begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}\right) = 2$. Thus, $\left\langle \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix} \right\rangle$ has exactly 2 elements.

Problem 6:
Let G be an abelian group and let $a, b \in G$. If $\operatorname{ord}(a) = 12$ and $\operatorname{ord}(b) = 18$, then what does $\operatorname{ord}(ab)$ equal?

Solution:

Let $n \coloneqq \operatorname{ord}(ab)$. By definition of order, we know that $(ab)^n = 1$. Since G is abelian, we know by the result of **Exercise 1 (c)** that:

$$(ab)^n = a^n b^n$$

So, since $\operatorname{ord}(a) = 12$ and $\operatorname{ord}(b) = 18$, and the lowest common multiple of 12 and 18 is 36, we know that 36 is the smallest positive integer satisfying:

$$\begin{aligned}
(ab)^{36} &= a^{36}b^{36} && \text{(Since } G \text{ is abelian)} \\
&= (a^{12})^3(b^{18})^2 && \text{(By exponent laws)} \\
&= (1)^3(1)^2 && \text{(Since } \operatorname{ord}(a) = 12 \text{ and } \operatorname{ord}(b) = 18\text{)} \\
&= (1)(1) && \text{(Inductively by definition of the identity)} \\
&= 1 && \text{(By definition of the identity)}
\end{aligned}$$